

Question bank

UNIT-1

1. What do you mean by the 5Vs characteristics of Big Data? Where does high volume of data get generated? Analyze the features of 3Vs in any Big Data applications
2. Explain the following terminologies:
 - a) In-Memory Analytics
 - b) In Database Processing
 - c) Symmetric Multiprocessor Systems
 - d) Massively Parallel Processing
 - e) Difference between Parallel and Distributed Systems
 - f) Shared nothing architecture.
3. Identify the following techniques and systems
 - a. The technology helps query data that resides in a computer's RAM rather than data stored in physical disks
 - b. A coordinated processing of a program by multiple processors, each working on different parts of the program and using its own operating system and memory
 - c. A collection of independent computers that appear to its users as a single coherent system
 - d. Single common memory shared among two or more processors
 - e. Database system that is tightly coupled, which has access to a common memory.
4. Explain the structured, unstructured and semi-structured data. And give examples for each. Classify the following as structured/semi/unstructured data.
5. What is Bigdata?. Why bigdata. Give the difference between traditional business intelligence and big data.
6. Explain veracity, validity, volatility, and variability. Explain the classification of analytics and the top challenges of big data.
7. What is big data analytics? What big data analytics is'nt?
8. Explain CAP theorem.
9. Differences between SQL/NoSQL. Classify into different baskets of SQL / NoSQL.
10. What is NOSQL? Types of NOSQL? Why NewSQL and the advantages of NOSQL
11. Explain the Shared Nothing Architecture and compare it with Shared Memory and Shared Disk architectures in distributed systems using suitable examples.
12. Explain the CAP Theorem and describe how it applies to distributed banking systems. Explain the difference between CP and AP architectures during a network partition in banking applications.
13. Explain the i) Data Warehouse Environment ii) Hadoop Environment iii) Big data and Data Warehouse Coexistence

UNIT 2

1. Why is Hadoop required? What not RDBMS? Differences between RDBMS and Hadoop? What are the challenges of distributed computing? What are the key aspects of Hadoop?
2. Explain the Hadoop components and hadoop high level architecture
3. Explain the Hadoop distributed file system. (HDFS) with Namenode, Datanode, File Read, File write, and Hadoop default replica placement strategy With neat diagram.
4. Explain the different HDFS commands.
5. Analyze the objectives given, and write the appropriate commands
 - Objective 1: To get the list of directories and files at the root of HDFS
 - Objective 2: To get the list of complete directories and files of HDFS
 - Objective 3: To create a directory in HDFS

Objective 4: To copy a file from the local file system to HDFS

Objective 5: To remove a directory from HDFS

6. How does Hadoop architecture support the MapReduce model for processing big data?
Explain with reference to data flow.

Question bank

UNIT-2

1. Explain MapReduce Framework: Phases and Daemons. Apply the Map Reduce Model for the Word Count Process for the input text: “Hadoop Definitive Guide, Hadoop in Action, Map Reduce Design patterns.”
2. Give a brief on YARN architecture and the steps involved in it with a neat diagram.
3. Differentiate between resource allocation and application scheduling in Hadoop YARN. Explain, with suitable points, how YARN enables multiple data-processing frameworks such as MapReduce and Apache Spark to share the same cluster resources efficiently
4. What are the limitations of Hadoop 1.0, and how does Hadoop 2 YARN address them?

UNIT 3

5. Apply array operations in MongoDB by creating a collection with array fields, inserting elements into the array, and retrieving a subset of elements using array slicing in queries with suitable examples.
6. Apply the aggregate framework in MongoDB to a collection of documents where "Customers" has four documents. Filter out those documents where the "AccType" has a value other than "S". After the filter, it should be left with three documents where the "AccType": "S". It is then required to group the documents on the basis of CustID and sum up the "AccBal" for each unique "CustID" to perform operations such as grouping, filtering, and computing totals or averages. Demonstrate with suitable queries how aggregate functions can be used to summarize data.
7. Explain the process of replication and sharding with a neat diagram in mongo dB
8. Explain the features of Apache Cassandra. Write and describe each feature clearly with suitable examples if necessary
9. Apply the knowledge on Apache Cassandra and Design a scalable data storage solution for an application. The application needs to handle high write throughput and provide efficient querying capabilities for the following use cases:
 - i) Create Student-info with Roll_No, StudName, DoJ, Last-exam-percentage
 - ii) To view only those records where Roll_No column either has a value 1 or 2 or
 - iii) Update StudName column to David Sheen for the record where Roll_No has value =2.
 - iv) To Delete the column “Last-exam-percentage” from the Student-info table for the record where the Roll_No=2’
10. Apply the concept of tunable consistency in Apache Cassandra to a real-time application scenario. Demonstrate how you would choose appropriate Read and Write consistency levels for different use cases
11. Apply the MongoDB commands find(), save(), pretty(), and upsert in MongoDB to perform database operations on a given collection. Demonstrate their usage with suitable examples.
12. Create a database in MongoDB by name DSCE, create a collection Student with following fields StudRollNo, StudName, Grade, Hobbies and DoJ. Write queries to perform the following

- i) Insert 5 documents
- ii) To display student name and Grade of the students when StudRollno is 1.
- iii) Update the Hobbies field of Aryan David as 'Skating'
- iv) To display 3 document only

13. Design a scalable data storage solution for an application using MongoDB . The application needs to handle high write throughput and provide efficient querying capabilities for the following use cases: Using the above Student collection develop queries for the following objectives.

- i) Find all documents with student name starting with letter 'M'.
- ii) Find all documents with student name ending with letter 's'
- iii) Find all documents with student name containing letter 'e'
- iv) To Find all those documents where Hobbies is neither set to "Chess" nor "skating"
- v) To find all those documents where Hobbies is neither set to "Net Surfing" and StudName= "Hersch Gibbs"

12. List five commonly used CQL data types and explains the purpose of each in the context of a realistic application such as a user-profile or order-management system.

13. Differentiate between the concepts of a Keyspace and a Table in Cassandra, and explain how they relate to similar concepts in a traditional RDBMS.

14. With syntax write query in Cassandra to Delete a column, a Table and Database.

15. Illustrate why MongoDB is preferred over RDBMS, with its feature. And mention the JSON records representation.

16. Apply the MongoDB commands count(), limit(), sort(), and skip in MongoDB to perform database operations on a given collection

17. Give basic different operations of CRUD with RDBMS vs MongoDB

18. Give a brief on Read Consistency and Write Consistency levels in Cassandra.

19. What are Keyspaces in Cassandra . Write query in Cassandra to insert the data into the column family "Student_info"

20. What are collections in Cassandra. Which are the different types present. And write the query in cassandra to create a table "users " with an emails column . the type of this column emails is set.

Question bank

UNIT-4

1. Illustrate how Hive is used for log file analysis in big data environments.
2. Write a Spark program to:
 - i) Create an RDD from a text file using `textFile()`
 - ii) Apply a `filter()` transformation to extract lines containing a specific keyword (e.g., "error")
 - iii) Display the first matching line
3. Create a table `Student_Info` using collection data types, `RollNo` Int, `Name` String, two Subordinate names use Arrays and for three subject marks use MAP. Terminate the Row format by `,`(comma), Collection item by `:` and MAP keys by `!`
 - i) Load the data into to the table from local path
 - ii) Display `Stuent_Info` all records (atleast 3)
 - iii) To retrieve name and subordinates names
 - iv) To retrieve Name and Marks1
 - v) To retrieve Name and first subordinates name
4. Write a Spark program to:
 - i) Persist the RDD using `persist()`
 - ii) Perform multiple actions (`count()`, `first()`)
5. A company stores customer order data by order date and customer region and runs frequent queries to analyze total sales per region and per month, as well as joins with a customer's dimension table. Analyze whether to use: Only partitioning, Only bucketing and Both partitioning and bucketing
6. Use HQL to achieve the following objectives :
 - i) Create a database by name `STUDENTS` with comments and database properties.
 - ii) Query to display databases, Describe database and remove database.
 - iii) To create manage table named `STUDENT`.
 - iv) To create self-managed table named `EXT_STUDENT`.
 - v) To load data into table `EXT_STUDENT` from file name `student.tsv`
7. A company stores sales data by year and product Analyze whether to use: Only partitioning, Only bucketing or Both. Justify your design with an example.
8. Explain the architecture of Apache Hive.
9. Elucidate how Hive is used for log file analysis in big data environments.
10. Create a table `Student_Info` using collection data types, `RollNo` Int, `Name` String, two Subordinate names use Arrays and for three subject marks use MAP. Terminate the Row format by `,`(comma), Collection item by `:` and MAP keys by `!`
 - i) Load the data into to the table from local path
 - ii) Display `Student_Info` all records (atleast 3)
 - iii) To retrieve name and subordinates names
 - iv) To retrieve Name and Marks1
 - v) To retrieve Name and first subordinates name
14. Use HQL to achieve the following objectives :
 - i) Create a database by name `EMPLOYEE` with comments and database properties.
 - ii) Query to display databases, Describe database and remove database.
 - iii) To create table named `Department`.
 - iv) perform aggregates per department
 - v) Use join operation with department table & employee table to display employee details along with their department name.

16. Given a scenario where a data analyst wants to process multiple CSV files like sales_jan.csv, sales_feb.csv, sales_mar.csv stored in HDFS into a single Resilient Distributed Dataset (RDD) in Apache Spark, (i) design and write a Spark program to create an RDD from these CSV files (ii) apply transformation operations to clean and transform the data finally use an action operation to display or compute a meaningful result.
17. Illustrate the Spark unified stack components with a figure.

UNIT 5

18. Develop concepts of Classification and Regression using MLlib in Python.
19. Illustrate with neat diagram the typical steps in a machine learning pipeline.
20. Elucidate how different data types in machine learning affect the choice and behavior of learning algorithms also interpret how algorithms would preprocess that data.
21. Analyze how TF (Term Frequency) and IDF (Inverse Document Frequency) together determine the importance of words in a document. Evaluate the TF-IDF score for the following two documents: **D1**: “machine learning machine” **D2**: “data analysis”
22. Apply collaborative filtering using the ALS algorithm in Apache Spark MLlib to recommend a product for a user based on the following ratings dataset. Show the steps involved in prediction.
- | User | M1 | M2 | M3 |
|------|----|----|----|
| U1 | 4 | ? | 5 |
| U2 | ? | 3 | 2 |
23. Apply collaborative filtering using the ALS algorithm in Apache Spark MLlib to recommend a product for a user based on the following ratings dataset. Show the steps involved in prediction.
- | User | M1 | M2 | M3 |
|------|----|----|----|
| U1 | 5 | 2 | ? |
| U2 | 3 | ? | 4 |
| U3 | ? | 1 | 2 |
- . Develop a spam classification system using Apache Spark MLlib in Python